

Exploring the Security Issues of Real World Assets (RWA)

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Abstract

Having real-world assets (RWA) tokenized on blockchains has recently been intensively discussed and recognized by DeFi ecosystem and traditional financial institutions as the next growth frontier, with its total value locked (TVL) surpassing \$2.5 billion as of September 30, 2023. However, the operational mechanics and security issues of RWA remain largely uncharted. To fill this gap, we propose the ever-first research on its anatomy and security issues. We initially outline two strategies for tokenizing real-world assets, and then we analyze the potential security risks within six associated entities (e.g., investors and smart contracts). Furthermore, we conduct a measurement of 39 mainstream RWA projects and reveal that RWA projects are facing risk deficiencies in areas of on-chain investors, stablecoins, blockchain oracles, project teams, and off-chain assets. Our research proves potential risks and threats in the RWA area and offers guidance for the burgeoning RWA market.

CCS Concepts

• Security and privacy → Economics of security and privacy.

Keywords

Blockchain; Real world assets; Cyber Security

ACM Reference Format:

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1 Introduction

The concept of Real World Assets (RWA) in Decentralized Finance (DeFi) refers to digitizing and tokenizing real-world assets, e.g., real estate, bonds, etc. With off-chain assets brought onto the blockchain and on-chain tokens redeemed for offline investments, RWA introduces more possibilities and playstyles for DeFi. A common case is that investors, operating in regions beyond the U.S., might have the capability to acquire and retain tokenized representations

of U.S. Treasury bonds and subsequently benefit from the associated yields. By September 30, 2023, the total value locked (TVL) of RWA projects surpassed \$2.5B [67] while the tokenization of global illiquid assets is anticipated to culminate in a market size of approximately \$16T [57] in 2030, showing the immense potential of RWA.

A few sources have discussed the broader advantages and implications of RWA. For instance, the Federal Reserve [31] discussed the advantages of tokenization and assessed its financial impact. Werner [90] refers to the concept of bringing off-chain assets on-chain as financial derivatives, i.e., synthetic assets created under the DeFi protocols. While RWA has gradually attracted widespread attention in the industry [22, 74], academic research on its introduction and security issues remains relatively limited. Meanwhile, with the risks inside blockchain infrastructure and various decentralized applications (DApps) being widely explored, e.g., blockchain networks [61, 95, 98], smart contracts [64, 77, 98], oracles [24, 27, 65], NFT [41, 88], etc, the field of RWA remains unknown and overlooked. To fill this gap, we are the first to propose research on its anatomy and security issues.

Specifically, we first outline the two major RWA models, i.e., *using on-chain funds to invest in real-world assets*, and *directly tokenizing off-chain assets*. Next, we analyze the six key entities (the investor, stablecoin, blockchain oracle, smart contract, project team, off-chain asset) involved in RWA models and delve into associated risk details, about which later we evaluate the performance of 39 mainstream RWA projects [42, 75]. The analysis results reveal risk deficiencies within RWA industry, i.e.,

- Most RWA projects display notable shortcomings in user identification (KYC), user verification (2FA), user-fund investigation (AML).
- The risks of stablecoins & blockchain oracles are potentially introduced in RWA
- RWA projects typically release whitepapers or documents that lack substantial real-world attributes, such as legality statements. Additionally, these projects do not provide adequate risk statements, and their asset management reports are generally superficial.
- Value deviation and loan default are the problems for the RWA industry.

The main contributions of this research are:

- To the best of our knowledge, this is the first paper to reveal the anatomy, entities involved, and potential risk of the concept of Real World Assets (RWA), providing valuable guidance for the burgeoning market.

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- We propose a thorough risk assessment of the RWA area from aspects of both the off-chain and on-chain, including the investor, stablecoin, blockchain oracle, smart contract, project team, and off-chain asset.
- We conduct an empirical research on 39 mainstream RWA projects and establish a evaluation of their security issues. The seven findings suggest that RWA is worryingly facing many potential risks and threats.

2 Background

2.1 Blockchain

Blockchain [92] is a distributed, shared, and immutable ledger that stores and manages data in blocks, secured and ensured for integrity using cryptographic methods. **Bitcoin** [91] is the first blockchain-based cryptocurrency that allows peer-to-peer transactions without intermediaries or banks.

Ethereum [93] is the first blockchain platform to support smart contracts. **Smart contracts** [97] are computer programs that automatically execute predefined logic when specific conditions are met. Since a smart contract is deployed, its logic cannot be altered. So if the code of smart contracts isn't open-sourced and users are unable to view and verify its workings and potential vulnerabilities, developers could have backdoors integrated within contracts, allowing unexpected operations like unlimited token minting or stealing one's tokens. A **transaction** [96] in Ethereum represents the value / data transfer from one account to another, e.g., funds transferring or a contract function calling. Ethereum has two types of **accounts** [47], i.e., Externally Owned Account (EOA) and Contract Account (CA). EOAs can send transactions (including transfers and smart contract calls) but cannot have their own code logic. CAs are controlled by the code of a smart contract, executing based on their internal logic when receiving a transaction.

2.2 Decentralized Finance (DeFi)

Decentralized Finance [90] represents a blockchain-based system that encompasses financial services but bypasses centralized intermediaries such as brokers. The DeFi ecosystem consists of lots of token contracts (e.g., ERC-20 tokens [43], which follow norms that standardized rules and interfaces for assets transfer / approval), and financial protocols [98] such as lending platforms and token exchange based on Automated Market Makers (AMM) [66] that provide liquidity.

RWA is not a fully on-chain DeFi project. It involves the tokenization of real-world assets, such as real estate, commodities, or other physical items, and integrating them into the blockchain ecosystem. Unlike typical DeFi projects, which operate entirely within the blockchain, RWA projects bridge the gap between the physical and digital worlds. By doing so, they expand the scope of decentralized finance, offering new opportunities for investors and participants in both the digital and physical asset markets [67].

3 Anatomy of the RWA Model

In this section, we delve deeper into the RWA mechanism, identifying two predominant strategies.

3.1 On-chain to Off-chain Liquidity Model

We refer to the behavior of RWA projects using on-chain tokens for off-chain investments, and transferring the returns to on-chain investors in the form of token values, as the *On-chain to off-chain liquidity model*. As Figure 1 illustrates, **1)** investors invest stablecoins into the RWA projects, i.e., interacting (e.g., staking) with DeFi protocols. **2)** The project team utilizes the substantial funds in DeFi protocols for investment activities, e.g., extracting stablecoins, converting them into USD, and channeling these funds into tangible assets. **3)** The project team (hopefully) garners associated investment rewards. **4)** Through DeFi protocols, the project team provides investment earnings to on-chain investors through mechanisms like token value increment / distribution.

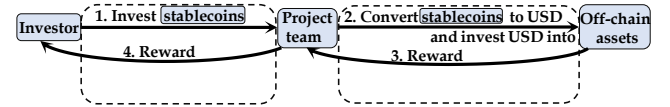


Figure 1: On-chain to off-chain liquidity model.

For example, here is how Matrixdock [11] (\$STBT) [63], a successful embodiment of RWA, aligns. **1)** Accredited investors interact with Matrixdock's token issuance platform using USDC (stablecoin), resulting in the minting of \$STBT into their wallets. This token offers users a decentralized gateway to access the returns from short-term US Treasury bonds (T-bonds). **2)** Matrixdock converts the USDC collected from investors who purchase \$STBT into USD suitable for real-world investments and then invests these funds in US T-bonds with maturities of 6 months or less, ensuring that each \$STBT is fully backed on a 1:1 basis by these T-bonds. **3)** Matrixdock receives returns from these T-bonds investments. **4)** Matrixdock channels the profits back to the on-chain investors, allowing them to earn an average annual percentage yield (APY) of 4.8% [29].

3.2 Off-chain Asset Tokenization Model

The *Off-chain asset tokenization model* refers to valuating & pricing real-world assets and directly issuing corresponding RWA tokens, offering investors an opportunity to purchase / hold / generate returns from these tokens. As Figure 2 illustrates, **1)** The project team first conducts a valuation analysis of the real-world asset, and with the asset is valued and priced, corresponding RWA tokens could be issued. **2)** On-chain investors then use stablecoins to invest in these RWA tokens such as purchasing. **3)** The project team extract stablecoins and convert them to USD, and better manage the off-chain assets such as asset custody. **4)** The project team earn profits that real-world asset generates (e.g., rental income, dividends). **5)** These profits are then distributed to on-chain investors as a reward.

We utilize RealT [73] (A real estate tokenization project) as an example to illustrate the process. **1)** In August 2023, RealT tokenized the property located at '4437 Lillibridge St, Detroit, MI 48214' [69] at the price of \$65,923.00 based on the details (e.g., 1,332 square feet with 3 bedrooms and 2 bathrooms, etc.) of this house. Ownership of this property is issued into 1,300 ERC-20 tokens¹ priced at \$50.71

¹0x969e3e55c8a4d9c4cd5e2660a2000ae12aad9e3e

($1300 * \$50.71 / \text{token} = \$65,923.00$). 2) By this means, investors use DAI [7] (stablecoin) to purchase the token. 3) RealT may extract DAIs and convert them to USD to manage the property, e.g., paying the 'maintenance expenses' and 'property taxes' [70]. 4) The project team would earn the rental income since listed properties on RealT are all rental properties. 5) RealT offers three ways for investors to gain rewards on a weekly basis, e.g., USDC distribution [71].

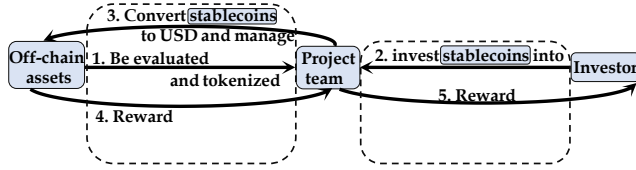


Figure 2: Off-chain asset tokenization model.

4 Entities Involved & Internal Risks

In this section, we introduce the common entities involved with two RWA models and survey their internal risk. The internal risks are not unique to any single model but are common challenges that both the *On-chain to off-chain liquidity model* and the *Off-chain asset tokenization model* may encounter in real-world applications.

4.1 Investors

Each on-chain investor can be considered as a blockchain account (e.g., EOA in Ethereum). Through their engagement with DeFi protocols and RWA tokens, they serve as the primary drivers for raising capital for off-chain investments, e.g., assets purchasing and financing loans.

Assuming that there are no restrictions for investors in place, i.e., they directly have interaction with DeFi protocols via their wallets, only their account addresses are recorded and traceable, without any personal information of the investors. However, transactions involving real-world assets have stricter requirements for investor identity verification. Take the aforementioned STBT [63], a token backed by the U.S. Treasury, as an example. Although STBT provides a simplified pathway for users to purchase bonds and enjoy related yields, the process itself of buying or redeeming the bonds in the real world has restrictions regarding the user's nationality, criminal record, investment history, etc [83]. Therefore, we have set up three risk factors to evaluate the potential risks that investors might bring, i.e.,

- Know Your Customer (KYC) [94], an essential procedure in the financial and business sectors, requiring companies to verify the identity of the customers they transact with, ensuring these customers are genuine and legitimate.
- Anti-Money Laundering (AML) [55], a set of regulations, laws, and procedures that aim to prevent criminals from disguising illegally obtained funds as legitimate income.
- Two-Factor Authentication (2FA) [56], a security measure that requires users to provide two different types of identity verification factors to prove their identity. 2FA has its limitations and can be vulnerable to certain attacks, such as SIM swapping [59]. However, it remains a crucial element in strengthening user authentication.

For each RWA project, we observe whether users are required to provide personal information such as identity proof, passport, address proof, etc., during registration or transactions, whether there is a clear AML statement or mechanisms for monitoring and reporting suspicious transaction patterns, and whether a dynamic verification code needs to be entered during the login process through methods like SMS or applications (e.g., Google Authenticator).

4.2 Stablecoins

Stablecoins [60] are cryptocurrencies pegged to traditional assets like the USD, offering a more stable mode of fund transfer and storage compared to other cryptocurrencies like Bitcoin or Ethereum. For example, a user pledging one USD to Circle (A FinTech Company) can mint one USDC (a mainstream stablecoin) [38].

Stablecoins are commonly used for staking, purchasing, or by project parties to exchange for dollars for off-chain investments in RWA projects. Although they are designed with stability in mind, their inherent risks inevitably intertwine with RWA. In STBT [63], the project converts on-chain funds into USD, then invests those dollars in US bonds, and subsequently redistributes the returns to users. This means that at certain moments, the project might hold a substantial amount of stablecoins. Should these stablecoins experience a sudden devaluation, the project faces the risk of significant asset depreciation. In another instance, within the lending scenario of Goldfinch [53], the company's settlement currency is USD converted from stablecoins. If the value of a stablecoin drops to near-zero, the project might need to consider holding other assets like ETH or BTC to meet the demands of the original lenders.

Furthermore, concerning stablecoins themselves, especially the centralized ones like USDT [85] and USDC [38], they heavily rely on issuing institutions or third parties to guarantee their value. If these entities face operational issues or their reputation gets tarnished, both the trust and value of the stablecoin might be jeopardized. The depeg of stablecoins has already happened multiple times [30, 84]. For instance, the value of UST (once the third largest stablecoin) plummeted to zero in May 2022 [26], serving as the tipping point that led to the bankruptcy of Three Arrows Capital, once one of the largest blockchain investment firms. Academically, researchers have pointed out the impact of the risk of devaluation of stablecoins, e.g., Eichengreen et al. [44] suggested regulating stablecoin issuers in a manner similar to banks, and they mentioned four risks inside stablecoins, including that payment risks are subject to devaluation if a company has receivables denominated in stablecoins.

4.3 Blockchain Oracles

The blockchain oracle [37] is a mechanism used to introduce external data into blockchains or smart contracts, enabling them to execute specific actions based on changes in external data. It allows smart contracts to interact with real-world data, such as prices, weather, match results, and more. For example, in a soccer lottery smart contract, specific logic needs to be executed based on real-world soccer scores, with blockchain oracle typically serving as an off-chain data provider.

However, introducing blockchain oracles inevitably comes with some risks [46], e.g., data source vulnerabilities, where malicious attackers may attempt to manipulate data sources for personal gain

or trigger undesired contract executions. Also, inadequate market coverage can lead to blockchain oracles incorrectly reporting asset prices [35] since attackers could manipulate the prices in those few markets the oracle refers to. In the past, oracles have been implicated in numerous security incidents, e.g., the first oracle security incident occurred in June 2019 [25], when a malfunctioning oracle reported incorrect prices to the blockchain, leading to over 37 million sETH being traded at low prices on Synthetix (a DeFi derivatives platform), involving nearly one billion USD. Therefore, when introducing blockchain oracles in RWA projects, a related risk emerges.

4.4 Smart Contracts

We primarily study token contracts that possess off-chain asset attributes (e.g., OUSG [49] is an ERC-20 token underpinned by an ETF consisting of short-term US governmental bonds), and focus on protocol contracts that execute financial services (e.g., the Goldfinch lending protocol allows on-chain users to lend their USDC to real businesses).

Audit firms typically examine and test smart contracts from a more professional perspective, including the business logic and standard vulnerabilities [79]. Simultaneously, audit reports lower the barrier for investors to read and inspect the source code. Considering the substantial integration of off-chain assets and on-chain tokens in the RWA process, ensuring the transparency, security, and stability of the smart contracts (including token contracts and DeFi protocols), is crucial. For instance, since 2023, Ondo Finance (one of the leading RWA projects [50]) has open-sourced its smart contracts on GitHub and has undergone three audits. Referring to the most recent report by Zokyo [51, 100], it revealed 11 vulnerabilities or coding irregularities within the project's contracts supporting RWA. For example, the report mentions that there's no validation in place for the transaction sender to be authorized to claim the deposit's excess in the specific contract. The sole requirement is undergoing KYC. Hence, anyone who has completed the KYC process could potentially claim the excess. Furthermore, Zokyo also tested the contract against 18 standard vulnerabilities, such as re-entrancy and unexpected ether balance [52, 80], to ensure its robustness.

Considering that auditing code requires a professional team with specialized knowledge and involves a significant amount of work, we adopt a lenient approach to assess the emphasis each RWA project places on the security of its project code. Specifically, we examine two risk factors: whether the smart contracts of each RWA project are open-sourced and whether they have been audited by third-party audit firms.

4.5 Project Teams

The project team refers to companies, teams, or DAOs (Decentralized Autonomous Organizations [89]) that champion and realize their RWA concept. Their main responsibilities include establishing company structures, tokenizing assets, managing and utilizing funds from on-chain investors, and ensuring proper risk mitigation, etc. RWA project teams are involved in a wide range of risks from a broad perspective, such as ensuring regulatory compliance, and maintaining transparency in asset management. Listing all these risks comprehensively can be quite challenging. However, we focus

more on the measures taken by these teams to ensure accountability to users throughout the project-building process. Specifically, we consider the following key elements:

- **Whitepaper / Document.** It is essential for the project team to openly disclose the project-building details such as its operational mechanisms, technical architecture, and team background. For example, past research suggested that technical information in the whitepaper can effectively signal the merit of a blockchain project [48]. Also, the existence of a whitepaper could serve as one of the identical factors to avoid 'Rug Pull' [68], where the project builders / developers disappear suddenly. Normally, the project team would release it or other descriptive documents before launching the project.
- **Investment / Custody disclosure.** Given that the project team oversees the custodianship of off-chain assets and substantial funds, they bear the responsibility to uphold investment transparency, entailing investment information like detailed asset description and proof of holdings, and investment specifics such as balance sheets. Otherwise, investors may face information asymmetry and are unable to fully comprehend how their funds are being utilized and the true financial status of the project.
- **Risk statement.** It is vital to elaborate on these risks of the blockchain networks, smart contracts' vulnerabilities, etc. Notably, as RWA introduces risks from real-world financial activities onto the blockchain, such as declines in government bond returns due to interest rate adjustments or real estate losses from natural disasters, the project team should ensure their risk disclosures align with financial product / service standards and provide a holistic view of potential risks to investors.

4.6 Off-chain Assets

The off-chain assets associated with RWA are quite diverse, encompassing U.S. treasury bonds, gold, silver, real estate, private credits, and carbon credits, etc. These assets hold immense economic value and also play vital roles socially and environmentally. U.S. Treasury bonds [82], as a type of fixed-income investment, have become an essential part of asset tokenization, with a market value exceeding 680M USD [28], and is regarded by most on-chain investors as 'risk-free'. Gold and silver are favored for their inflation-hedging characteristics and high liquidity. The private credit market is also a crucial sector [75], where funds extracted from the blockchain can be provided to different real-world enterprises.

Each category of assets has its unique characteristics, liquidity, and risk profile. The diversity makes the valuation of assets extremely complex, and it directly affects market transparency and the interests of investors. To ensure fairness and authenticity in valuation, we introduce a risk factor called 'value deviation', referring to the lack of market reference information for off-chain assets or a discrepancy between the valuation of off-chain assets and their value claimed on the blockchain. For instance, an uninformed investor might overpay for a token backed by real estate of limited actual value; or the on-chain promised returns might significantly differ from the actual earning of the off-chain assets. As a result,

we place emphasis on examining the consistency and discrepancy between the publicly quoted price of the underlying assets, third-party reference prices, the returns promised on blockchains, and any other value-related information.

Another manifestation of off-chain assets is credit. In the realm of credit, a common risk inside is the ‘loan default’, where the funds of on-chain investors are intermediated by the project team, lent to a third party, and might eventually vanish without a trace.

5 Analysis Results

In this section, as shown in Table 1, we present the empirical results on 39 mainstream RWA projects in terms of their security issues.²

5.1 Projects Investigated

We focus on 39 RWA projects sourced from RWA.XYZ [75], which provides industry-standard reference data for every tokenized real-world asset in one place. The projects we choose covers a wide range of real-world assets, including U.S. Treasury bonds, farmland, real estate, gold, silver, and more. Among these projects, 17 are based on the *Off-chain asset tokenization model*, while 22 follow the *On-chain to off-chain liquidity model*.

Entity	Risk factor / Detail	Employed	Not Employed	O / N / U
Investors	KYC	19 (48.72%)	19 (48.72%)	1 (2.56%)
	AML	15 (38.46%)	23 (58.97%)	1 (2.56%)
	2FA	7 (17.95%)	29 (74.36%)	3 (7.69%)
Stablecoins	Stablecoins	35 (89.74%)	1 (2.56%)	3 (7.69%)
Blockchain oracles	Blockchain oracles	17 (43.59%)	22 (56.41%)	0 (0.00%)
Smart contracts	Open-sourced	30 (76.92%)	9 (23.08%)	0 (0.00%)
	Audited	26 (66.67%)	12 (30.77%)	1 (2.56%)
Project teams	Whitepaper / Documents	31 (79.49%)	7 (17.95%)	1 (2.56%)
	Off-chain interaction statement	11 (28.21%)	27 (69.23%)	1 (2.56%)
	Legality statement	8 (20.51%)	30 (76.92%)	1 (2.56%)
	Restricted area	13 (33.33%)	25 (64.10%)	1 (2.56%)
	Risk statement	14 (35.90%)	25 (64.10%)	0 (0.00%)
	(Asset-related) risk statement	12 (30.77%)	27 (69.23%)	0 (0.00%)
	Investment / Custody disclosure	12 (30.77%)	22 (56.41%)	5 (12.82%)
Off-chain assets	Value deviation	20 (51.28%)	0 (0.00%)	19 (48.72%)

Table 1: The measurement result of RWA projects. Notably, ‘O / N / U’ stands for ‘Optional / Not applicable / Unknown’, representing the total of these categories, which are not within the scope of our detailed discussion.

5.2 Investors: KYC / AML / 2FA

In our investigation of RWA, we find significant shortcomings in its KYC, AML, and especially 2FA compliance measures. As Table 1 illustrates, less than half successfully implement KYC measures (48.72%) and AML measures (38.46%). Even more concerning, only seven projects, representing 17.95%, adopt 2FA measures. Alarmingly, although RealT [73] claims to have a KYC process, in our practical test, we can easily bypass the video verification using a book photo instead of an ID card, and then have immediate interaction with the platform, exposing a substantial security loophole. Furthermore, while projects like Stream finance [16], Solid world [15], and Jia [9] claim to support KYC and conduct user background checks, they haven’t publicly declared any AML strategies or measures. Notably, even though Polytrade [14] emphasizes in its whitepaper

²Our results are publicly available at https://github.com/chenshj73/defiCCS_RealWorldAssets

and related interviews that it will adopt KYC and AML measures, in practice, users simply need to register on the platform to connect their wallets and start trading. Also, projects like stUSDT [17] and Bloom Garden [3] perform the worst, lacking measures in KYC, AML, and 2FA altogether.

Finding 1: *The vast majority of RWA projects display notable shortcomings in user identification, verification, and user-fund investigation.*

5.3 Stablecoins

We assess each RWA project to determine whether mainstream stablecoins [40] (USDC / USDT / DAI / TUSD / BUSD) serve as its medium of payment or store of value and analyze whether the project has any reserve funds or mechanisms in place to address significant stablecoin fluctuations. It turns out that the vast majority of RWA projects (89.74%, see Tab. 1) employ stablecoins for transactions or as reserves and none of them are prepared for the associated risks. This not only demonstrates that stablecoins have become the mainstream mechanism to bridge on-chain investors with real-world assets, but it also suggests that most RWA projects might be influenced by the value of stablecoins. Among them, USDC is the most preferred stablecoin, being used in 80% of RWA projects that incorporate stablecoins. However, in 2023, due to the bankruptcy of Silicon Valley Bank, USDC’s \$3.3 billion in reserves suffered a significant hit, leading to a severe disruption in its pegging relationship with the US dollar.

Finding 2: *Most RWA projects inherently introduce risks associated with stablecoins.*

5.4 Blockchain Oracles

Table 1 shows that almost half of the RWA projects have adopted the blockchain oracles. Some projects like Yieldteq [19] use oracles to track the real-time price of their native token USTY; while projects like Clearpool [6] adjust economic model parameters by integrating oracles into their DAO structure. However, it is more common for RWA projects to employ the Proof of Reserve (PoR) [36] service from oracles, such as Backfi [2], CACHE.Gold [4], and Matrixdock [11] and etc. PoR provides collateralization ratios for any off-chain assets, and is increasingly used to ensure the safety of the minting, redemption, and destruction processes of wrapped assets. For instance, OpenEden T-Bill [13] allows a decentralized oracle node network to connect with vault’s off-chain custody account, after which chainlink provides a daily snapshot of vault’s net asset value (NAV). Furthermore, some RWA projects have already recognized the potential risks introduced by oracles. For example, Elyfi [8, 45] points out in its documentation that although data from oracles reflected on the blockchain is typically trustworthy, there is still a risk of errors or malicious alterations. To address this issue, in the context of tangible asset-backed lending, they allow a large number of participants to be involved in the formation of loan contracts, aiming to minimize the risk of data tampering.

Finding 3: *Over half of the RWA projects might have introduced risks associated with oracles, with Proof of Reserve adopted most.*

5.5 Open-sourced / Audited Smart Contracts

Even though some projects opt not to go open-source, or their code has not been audited (e.g., RealT [72] has explicitly stated

that its code will not be audited, shifting this risk onto its users), it is evident that the vast majority of RWA projects lean towards adopting an open-source (76.92%) or audit (66.67%) approach (See Tab. 1), indicating a trend that most RWA projects adhere to the standard practices of the blockchain industry. Also, more projects have already begun auditing their code, e.g., the code of Lofty AI [10, 32], stUSDT [33] and TProtocol [18, 34] is currently under review. Thanks to the mature code security processes and systems in the blockchain, researchers might need to focus more on the security aspects of off-chain elements in the tokenization process.

Finding 4: Most RWA projects adhere to the standard practices of the blockchain industry by conduct open-sourcing / auditing.

5.6 Project Teams

5.6.1 Project team: Whitepaper / Document. 1) As shown in Table 1, the majority of RWA projects (79.49%) adhere to secure practices of blockchain projects by releasing their whitepapers, providing a clear and transparent overview for investors. However, there are still 7 projects (17.95%) that exhibit a deficiency in this crucial aspect of security by lacking this vital document or without any project explanation. Take Stream Finance [16] as an example, although it has issued RWA tokens, the absence of corresponding descriptive documentation and unestablished modules within the project (e.g., lending pools) could expose investors to unforeseen risks.

2) Off-chain interaction statement. While most existing whitepapers primarily focus on the interpretation of DeFi protocols, detailed elucidation of economic models, and user guides, they often neglect the description of the real-world activities, potentially hindering investors from fully understanding how project parties interact with off-chain entities and how on-chain funds are invested and managed, etc. Ondo Finance [50] presents a positive example by explicitly explaining the flow and usage of OUSG (A token backed by T-Bills), i.e., [49] while Coinbase Custody (off-chain entity)'s fund account manages users' USDC, Ondo IM (off-chain entity) is responsible for converting the account's USDC into dollars and purchasing ETFs at Clear Street (financial infrastructure provider). When the ETF pays dividends, Ondo IM will use the dividend income to purchase more of the same ETF and eventually distribute profit on blockchains.

3) Legality statement & Restricted area. As shown in Table 1, most projects (76.92%) exhibit a significant weakness in the description of legal compliance in their whitepapers or other explanatory documents, i.e., which country's or region's laws and regulations support RWA, posing legal risks to investors. According to public information, only the Hong Kong Monetary Authority (HKMA), has officially endorsed an RWA project (Evergreen) using traditional securities law for the first time [20]. Additionally, projects seldom (33.33%) publicly issue warnings or restrict users or investments from different countries or regions. Among them, investors from the United States receive the most warnings, followed by Mainland China, reflecting that RWA projects might fail to gain recognition and participation from investors in the two global superpowers.

Finding 5: RWA projects uphold the good practice of releasing whitepapers, but many fall short in detailing real-world activities and ensuring legal compliance.

5.6.2 Project team: Risk statement. We find that many RWA projects (64.10%, see Tab. 1) do not adequately explain their potential risks,

especially when excluding the disclaimers provided. What's more concerning is that these projects (69.23%, see Tab. 1) do not explicitly disclose the risks associated with off-chain assets, which could expose investors to real-world risks without key information. For instance, while CitaDAO [5, 39] elaborates on the risks related to blockchain and front-end interactions, it overlooks the risks of real estate in the financial market. In contrast, OpenEden T-Bills [13, 86] offers a more comprehensive risk disclosure, detailing the risks associated with its investment products, including interest rate, market liquidity, and U.S. sovereign credit risks. Such thorough risk descriptions not only help investors make more informed decisions but also enhance the transparency and credibility of the project.

Finding 6: Most RWA projects do not adequately disclose potential risks, especially those related to off-chain assets.

5.6.3 Project team: Investment / Custody disclosure. Only a few RWA projects (30.77%) transparently display the details of their off-chain investments and asset custody in Table 1, and the information disclosed often remains superficial. If we measure each project against the typical standards of financial reporting (i.e., the balance sheet, cash flow statement, and income statement), not a single project currently meets these criteria in full. Based on this, we have identified some issues present in the disclosures of RWA projects.

- Unauthentic disclosure, e.g., the Agatobwe [1] project uses three cameras to show investors the real-time status of off-chain assets, but in reality, it's just displaying three looped video clips.
- Overly brief information without concrete evidence, e.g., Lofty Ai [10] sporadically provides short textual updates about the property status, like whether tenants are paying rent on time, but doesn't offer any corroborative documents or third-party verification.
- Lack of consistent and ongoing updates, e.g., the NAV (Net Asset Value) report from OpenEden T-Bills [13, 87] has been delayed by a month (as of October 8th, 2023), and past reports are not accessible to investors.

Finding 7: RWA projects provide superficial and potentially misleading information on asset management disclosures.

5.7 Off-chain Assets

In Table 1, over half of RWA products (51.28%) may exhibit value deviation, and for the remaining ones (48.72%), investors' judgment becomes murkier due to the lack of real-world reference data, e.g., in private-credit-backed tokens, investors cannot discern how borrowers utilize on-chain funds, making it challenging to accurately gauge the returns on their investments. Also for this type of reason, we could not be able to judge whether there might be value deviation.

Real-world assets mainly fall into two categories, national / government bonds (33.33%) and real estate (20.51%). The value of these assets can typically be tracked. Nowadays, several online trading / investment platforms offer real-time tracking of bond yield rates, allowing investors to determine if there's a significant value deviation between their on-chain investment and the actual market. Taking the around 5% APY (Annual Percentage Yield) of US Treasury bonds as an example, we analyze 9 relevant RWA projects and find their APYs range between 3.8% - 5.37%, which seems reasonable

considering intermediary costs like bank and dealer fees. However, real estate values are influenced by various factors, such as location, property age, and condition. Currently, aside from real estate brokerage and data analysis firms, there isn't a universally recognized property price index. Thus, significant price deviations might occur when properties are tokenized, e.g., certain substandard properties might be overvalued. We reference Zillow [99] (the largest real estate data company in the US), and investigate 345 properties from RealT [73] (the largest real estate tokenization project). As shown in Figure 3, the findings reveal that 60.58% of the properties lack reference values, and the remaining properties have an average discrepancy of 24.61% compared to Zillow evaluations, suggesting possible overvaluation by RealT.

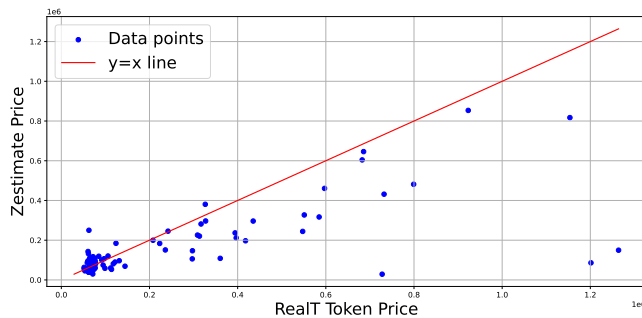


Figure 3: Value deviation between RealT token and Zestimate.

Moreover, the value of gold can also be tracked. For instance, the LBMA [58] Gold Price is a widely cited gold pricing standard. MeldGold [12], guarantees that each MACU (the gold-backed token) equates to the value of one gram of gold. However, as illustrated in Figure 4, when comparing the historical prices of MACU from November 18, 2020, to September 24, 2023, with the LBMA, we discover that since the latter half of 2022, MACU's price has decoupled from the gold price. We speculate that this indicates an arbitrage opportunity that people have not yet noticed.

Some RWA projects, concentrated in the credit sector, involve substantial funds. For instance, as of October 4, 2023, the RWA assets within the MakerDAO protocol reached \$3.1 billion [62]. However, even these RWA protocols (most of them are lending ones) cannot completely avoid the risk of loan defaults based on public security incidents. Taking Goldfinch [53] as an example, this project raises funds for off-chain operating entities (through tokens pledged by users) and provides users with part of the profits and loan interest from these companies. However, in August 2023, the Kenyan motorcycle company Tugende defaulted on a \$5 million loan [23], accounting for about 4% of Goldfinch's total value locked (TVL), indicating that while users can enjoy off-chain benefits, they might lose all their principal. Although loan defaults are a common adverse phenomenon in financial activities, the funds often come from the tokens of users and are converted into dollars, making the risk ultimately fall on on-chain investors. As of October 12, 2023, we find that there are 48 loan default events listed on the rwa.xyz [75]. Additionally, we find that at least seven loan default events have occurred through Twitter (X.com) crawlers and public investment reports, accounting for a total amount of \$106.34 million, including projects like MakerDAO [76, 81], Goldfinch [23], and TrueFi [78].

Finding 8: Value deviation and loan default are challenges that RWA projects may encounter in terms of off-chain assets.

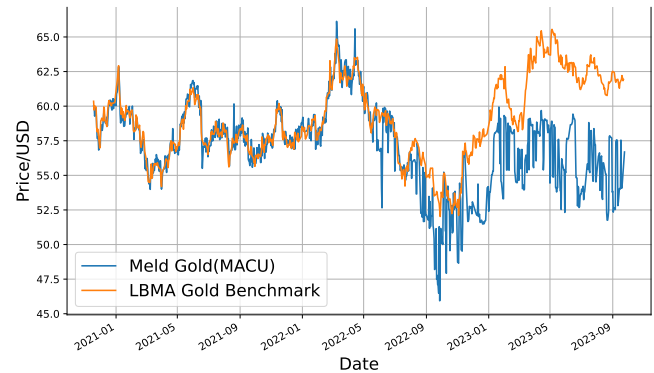


Figure 4: The Price deviation of the MACU token and the LBMA gold benchmark from Nov. 18, 2020 to Sep. 24, 2023.

6 Discussion

In this section, we discuss the use of Special Purpose Vehicles (SPVs) in RWA projects and provide a case study of the stUSDT project, analyzing associated limitations in our assessment framework.

6.1 Special Purpose Vehicle

Beyond the previously discussed RWA risks, the implementation of new RWA models introduces additional risks related to corporate structure and regulatory aspects. For instance, RWA projects often use Special Purpose Vehicle (SPV). An SPV is a legal entity specifically established for a particular purpose and is often used as a tool in financial activities such as structured financing and securitization [54]. Through the SPV structure, RWA can effectively isolate bankruptcy and liability risks between on-chain funds and off-chain assets. Take Goldfinch [53] as an example. If borrowers A, B, and C each use tangible assets as collateral to borrow funds from on-chain investors, Goldfinch would create an individual SPV for the lending pool respectively, ensuring three lending activities remain distinct and unaffected by one another. Furthermore, if a lending pool defaults or encounters other risks, it won't spill over to the main Goldfinch entity.

While SPVs offer a certain degree of bankruptcy and liability risk isolation for RWA projects, the level of their public disclosure and registration often remains ambiguous. For instance, RealT [73] claims that they have established a dedicated SPV for every property. However, it is difficult for us to locate every specific name associated with the SPVs of RealT. Also, the confirmation of registration details is necessary. Hence, we verify whether they've declared the use of SPVs and have publicly disclosed the SPV names. Following that, we delve deeper by accessing company registration websites from various regions to validate these claims. We find that only 38.46% of RWA projects have publicly stated that they adopt the SPV corporate structure to achieve risk isolation for on-chain assets. However, among these companies that have adopted SPV, only three (Ondo Finance [50], Loftly Ai [10] and OpenEden T-Bills [13]) have openly disclosed the name of their SPVs and have related registered records. While we understand that on-chain investors

may not pay much attention to real-world legal entities, a transparent corporate structure can help investors clearly understand which legal entity is managing their assets, thereby boosting their investment confidence.

6.2 Case Study: stUSDT

stUSDT [17] is the first RWA project introduced on the TRON Blockchain, where users are allowed to stake USDT to acquire stUSDT. The stUSDT project team will further convert the staked USDT to USD and invest it in short-term government bonds. When users redeem, they receive the principal along with an approximately 4% APY return. However, there are five risk factors of stUSDT under our assessment framework. **1)** The project does not release its whitepaper, and the official documentation only covers limited information like user guidance, and the existing documents do not align with actual operational mechanisms. For example, the promised DAO has not been released and the way they claimed to invest transparently has not been conducted. Meanwhile, stUSDT does not set any restrictions on countries or regions within its documents. **2)** The investment disclosure is too brief and poor. As shown in 'Investment Date (2): July 05, 2023, Maturity Date (2): October 05, 2023', stUSDT only reveals the date for tracking investment products, without details such as type, amount, returns, etc, and it is not daily updated and verified by third-party institution. **3)** stUSDT doesn't require any KYC / AML / 2FA processes for users to complete all interactions, meaning that they are totally unaware of the investors' identities and the sources of their funds. **4)** stUSDT does not use the SPV structure to manage and invest on-chain funds in the real world. We also cannot find any company entity related to stUSDT, which might suggest that stUSDT is still anchored to the native TRON parent company. **5)** The project's source code does not undergo third-party auditing and has centralized functions like stopping token minting as we could easily find.

6.3 Limitations

6.3.1 Human-based Assessment. In the process of manually retrieving information, there is a potential risk of misjudgment due to human factors. To ensure the reliability of our research, we adopt a relatively lenient assessment standard. Taking risk statements as an example, different RWA projects vary in the length and depth of their risk descriptions, where some might be concise, and others are more detailed. Our evaluation criterion is that as long as the project dedicates a specific section to risk description in its official documentation, whitepaper, or user interface, it is deemed to meet the assessment standard. Despite our lenient and cautious approach, the measurement results still reveal that many RWA projects' strategies in addressing multiple risk factors remain less than satisfactory.

6.3.2 Unknown information. Some projects require us to contact the project teams for account registration or relevant information. However, there are times when we don't receive a timely response from them. Additionally, some crucial web links (such as whitepapers or audit reports) have become invalid, or the project teams do not fully disclose information. For content that we cannot access, we mark it as 'Unknown'.

6.3.3 Time-sensitive results. Our results are time-sensitive, specifically up to October 12, 2023. The industry environment and data

may change over time, and we advise readers to consider the timeliness of the data when reading and referencing our work. However, we hope these results will provide useful guidance for the RWA industry and subsequent related work.

6.4 Ethics

We have performed a rigorous and detailed examination of multiple companies within the RWA industry. It is imperative to state that we hold no financial interests or affiliations with any of the entities analyzed. Our sole intention is to provide an unbiased and thorough assessment of the industry, contributing valuable insights and references for further study. All evaluations and conclusions are derived from publicly accessible data and conducted through independent research methods, ensuring the accuracy and transparency of the information presented.

7 Related Work

The security issues of RWA was rarely mentioned in previous studies. Werner et al. [90] referred to it as a derivative of synthetic assets in the world of DeFi, capable of replicating on-chain assets on the blockchain. The Federal Reserve [31] conducted research on the financial impact of tokenization, revealing that the total value locked (TVL) of RWA has been continuously growing compared to the entire DeFi market since July 2021. Baltais et al. [21] estimate that tokenizing real-world assets could generate global annual economic gains of approximately \$2.4 trillion due to increased efficiencies, with realistic gains between \$31 billion and \$130 billion by 2030, particularly benefiting the foreign exchange market. In contrast, researchers have focused more on the on-chain aspect of RWA. For example, Zhou et al. [98] have proposed a systematic measurement and evaluation framework for DeFi attack incidents. Das et al. [41] primarily revealed various unethical practices from both user and exchange platform perspectives in the NFT (Non-Fungible Token) space, such as tampering with token metadata and wash trading. Eskandari et al. [46] discussed the market manipulation risks associated with blockchain oracle design principles and related security events. Although RWA is a broader concept, to the best of our knowledge, this is the first paper to study and measure the security issues associated with RWA.

8 Conclusion and Future Work

We conduct the first study on the security issues of RWA. In particular, we provide an in-depth summary of the two major RWA models and delve into their six common entities and internal risks. We then perform a comprehensive evaluation of 39 mainstream RWA projects on the market, concluding eight findings. Apart from the finding that most RWA projects have their code open-sourced and audited, the results demonstrate RWA projects are facing risk deficiencies in areas like on-chain investors, stablecoins involved, blockchain oracles introduced, project teams, and off-chain assets.

In the future, we aim to conduct a deeper analysis of the security risks in RWA projects by developing more refined risk assessment models for different entities and risk types.

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